Algorithms 2026: Assignment-4

Due date: November 29, 2026

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1. Suppose you are given a stream of m elements, where each element belongs to the set $\{1, 2, \ldots, n\}$. Design a deterministic streaming algorithm that uses $O(\log n + \log m)$ bits of space and returns an element that appears strictly more than $m/2$ times, if such an element exists. If no such element exists, the algorithm's output may be arbitrary.

2. Suppose you are given a stream of m elements, where each element belongs to the set $\{1, 2, \ldots, n\}$. Design a randomized streaming algorithm that uses $O(\log m + \log n)$ bits of space and samples an element with probability proportional to its frequency in the stream. Specifically, an element that appears f times should be returned with probability f/m.

3. Given an undirected graph G, an edge colouring is an assignment of colours to the edges of G such that no two edges incident to the same vertex get the same colour. The edge colouring problem asks for an edge colouring using the minimum number of colours. This is a hard problem to solve. Here we are interested in the *online edge colouring* problem where the vertex set V of the graph G is fixed and the edges in the graph are presented to us in an online manner, one after another.

   As each edge e is specified, our algorithm must assign this edge e a colour and this colour of e cannot be changed henceforth. While the offline edge colouring of G knows all the edges of E while deciding on their colours, the online edge colouring algorithm has to decide on the colour of each edge e without any knowledge of the future edges. Design a 2-competitive algorithm for the online edge colouring problem.

4. Recall the online paging problem: the cache size is k. Suppose the total number of distinct items in memory is $k + 1$. Show that the competitive ratio of the Marker algorithm is H_k.

5. Recall the set cover problem: we are given a universe U along with a collection $\mathcal{C}$ of subsets of U such that $\bigcup_{S \in \mathcal{C}} S = U$; moreover, each set S has a non-negative cost $c(S)$. The problem is to find a min-cost subcollection $\mathcal{C}' \subseteq \mathcal{C}$ whose union is U.

   Consider the following LP-relaxation of the problem.

$$\begin{aligned} \text{minimize} \quad & \sum_{S \in \mathcal{C}} c(S)\, x_S \\ \text{subject to} \quad & \sum_{S : u \in S} x_S \;\geq\; 1 \quad \text{for } u \in U \\ & x_S \;\geq\; 0 \quad \text{for } S \in \mathcal{C} \end{aligned}$$

   Let $(x^*_S : S \in \mathcal{C})$ be an optimal solution to the above LP. Suppose the collection $\mathcal{C}$ satisfies the following additional condition: *every element of U appears in at most f sets in $\mathcal{C}$*. Consider the following rounding algorithm: if $x^*_S \geq 1/f$ then include S in our subcollection.

   Show that such a subcollection is indeed a valid set cover of size at most $f \cdot \mathsf{OPT}$, where OPT is the cost of the min-cost set cover.

6. Consider the following rounding rule for the above problem: if $x^*_S > 0$, then include S in our subcollection. Show that such a subcollection is also a valid set cover of cost at most $f \cdot \mathsf{OPT}$.

   (*Hint:* Write the dual and use complementary slackness.)

7. Consider any bipartite graph $G = (A \cup B, E)$. By König-Egerváry theorem, we know that the size of a minimum vertex cover in G is the same as the size of a maximum matching. So we can determine the cardinality of a minimum vertex cover in G in polynomial time by running a maximum matching algorithm. Suppose we want to *find* a minimum vertex cover C, i.e., the set C and not just the value of $|C|$. Show a polynomial time algorithm for this problem.

8. We wish to solve the minimum vertex cover problem on input instances that consist of a graph G, together with a valid vertex colouring of G with 3 colours. Give a 4/3-approximation algorithm for the minimum vertex cover problem on such instances.

   (*Hint:* You can use the fact that we can compute a minimum cardinality vertex cover in a bipartite graph in polynomial time.)